

Advanced Programming 2025-2026 Assignment #2: Deadline: 30/4/2026 Second semester

Develop a console-based Tic-Tac-Toe game in Python using:

- Object-Oriented Programming (OOP)
- Data Structures (lists and/or dictionaries)

Create a two-player Tic-Tac-Toe game where:

Players take turns entering their moves

The board updates after each move

The game detects a winner or a draw

Requirements

1. Use Object-Oriented Programming (OOP)

Your program must include at least two classes, such as:

Player → stores player name and symbol (X or O)

Game → controls the game logic

You must use:

`__init__` constructor

At least 2 methods per class

Attributes inside classes

2. Use Data Structures

You must use:

List → to represent the game board (e.g., 3x3 grid)

Example:

```
self.board = [' ' for _ in range(9)]
```

(Optional) You may use:

Dictionary → mapping positions (1–9) to board indices

3. Game Features

Your game must include:

Display the board after each move

Alternating turns between players

Prevent overwriting an occupied cell

Detect winning conditions:

Rows

Columns

Diagonals

Detect a draw

Option to replay the game

4. Functional Requirements

Your Game class should include methods such as:

display_board()

make_move(position, player)

check_winner()

is_draw()

switch_player()

Example Game Output

Player 1 (X), choose position (1-9): 5

```
| |  
-----  
| X |  
-----  
| |
```

Player 2 (O), choose position (1-9): 1

...

At the end of your program, include a short printed message describing any challenges you faced during development and whether you sought external help.

Instructions: submit your files via Moodle elearning. (you can use your github url)

Remark: Copy Paste is not allowed. And any two assignments with a similar solution, zero marks will be given for both.